Lab 01
House Price Prediction using Linear Regression

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Github link: **https://github.com/Harshita2011/House_Price_Prediction**

Contents

1. Executive Summary

This study developed and evaluated regression models for predicting residential property value on three publicly available datasets of increasing complexity: **Ames Housing** (1,460 individual home sales in Ames, Iowa; target `SalePrice` in USD), **California Housing** (20,640 census block groups from the 1990 U.S. Census; target median district value in USD 100,000s), and **UCI Real Estate Valuation** (414 transactions in New Taipei City, Taiwan; target price per unit area in 10,000 NTD per ping). Seven models — Linear Regression, Ridge, Lasso, Elastic Net, Decision Tree, Random Forest, and Gradient Boosting — were compared inside identical leakage-safe scikit-learn pipelines, with 5-fold cross-validation on the training split used for model selection and a single held-out 20% test split used once for final evidence.

Ensemble methods achieved the best generalization on all three datasets. On Ames, the selected **Gradient Boosting** model reached a test MAE of **$16,556**, RMSE of **$26,504**, and RZ of **0.908** — a 69.8% RMSE reduction relative to a mean-prediction baseline. On California, **Random Forest** reached MAE 0.327, RMSE 0.505 (USD 100k), and RZ 0.805 (−55.9% vs. baseline). On UCI, **Random Forest** reached MAE 3.913, RMSE 5.669, and RZ 0.808 (−56.2% vs. baseline). The strongest predictive signals were living area, overall quality, and neighborhood (Ames); median income and geography (California); and MRT-station distance and convenience-store count (UCI). The multiple linear regression baseline remained surprisingly competitive on Ames (test RZ 0.887), confirming its value as an interpretability anchor, while its collapse on California (RZ 0.576 vs. 0.805) quantifies the cost of ignoring nonlinear structure.

The models are suitable as decision-support estimates within their original data coverage. They should **not** be used for mortgage underwriting, taxation, or other high-stakes decisions without current local data, uncertainty estimates, segment-level validation, and human review.

2. Business Problem

A real-estate analytics team needs a repeatable method for estimating the market value of residential property from measurable attributes — size, age, quality, location, and nearby amenities. The learning task is supervised regression: estimate a function $\hat{y} = f(x)$ mapping property attributes to a continuous price target, evaluated on observations not used during fitting.

2.1 Stakeholders and error costs

- **Buyers and sellers** use estimates for price discovery; large errors distort affordability decisions.
- **Banks and mortgage firms** use valuation checks for collateral risk; systematic over-valuation increases credit risk, so RMSE (which penalizes large errors) is the primary metric alongside MAE.
- **Developers and investors** use estimates for screening; bias across locations can misallocate capital.
- **Public agencies** require transparent, consistent mass appraisal — favoring interpretable baselines as a reference.

2.2 Prediction units and success criteria

The three targets are deliberately different quantities and are never compared directly: Ames predicts an **individual sale price in USD**; California predicts a **median block-group value** in USD 100,000s (not an individual house price); UCI predicts **price per unit area** in 10,000 NTD per ping (not total price). Success criteria: (i) materially lower CV/test RMSE and MAE than a naive mean-prediction baseline; (ii) an acceptable gap between cross-validation and test performance; (iii) no unexplained severe residual pattern or leakage; (iv) errors expressed in meaningful currency units.

3. Dataset Description

Table 1 summarizes the three datasets. They form a progression from a small clean problem (UCI) through a large numeric problem (California) to a realistic high-dimensional mixed-type problem (Ames), which is the primary extended project.

Table 1. Dataset summary

Dataset	Rows	Predictors	Types	Target and unit	Source / coverage
Ames Housing	1,460	80 (37 num, 43 cat, excl. Id)	Mixed	SalePrice, USD	Ames, Iowa residential sales, 2006–2010 (De Cock, 2011)[1]
California Housing	20,640	8 numeric	Numeric	MedHouseVal, USD 100k (block-group median)	1990 U.S. Census, via scikit-learn[2][3]
UCI Real Estate	414	6 numeric	Numeric	Price per unit area, 10k NTD/ping	New Taipei City, Taiwan, 2012–2013 (UCI ID 477)[4]

Data-quality audit findings. Ames contains no duplicate rows, but 19 of 81 columns have missing values; the highest rates are structural absence codes rather than unknown data (e.g., `PoolQC` 99.5% missing means "no pool", `Alley` 93.8% means "no alley access") and were treated as a distinct category rather than blindly imputed. The target is strongly right-skewed (skew 1.88; mean \$180,921 vs. median \$163,000; range \$34,900–\$755,000). California has no missing values but a censored target (median values capped at \$500,001) and block-group aggregation. UCI is small, complete, and clean — six numeric predictors with no missing values.

4. Methodology

4.1 Experimental protocol

All three datasets were processed with the identical protocol, making cross-dataset comparison fair:

- **Holdout design.** 80/20 random train–test split with `random_state=42` before any learned preprocessing (Ames: 1,168/292; California: 16,512/4,128; UCI: 331/83).
- **Leakage-safe preprocessing.** A `ColumnTransformer` inside a scikit-learn `Pipeline` : median imputation + standard scaling for numeric features; most-frequent imputation + one-hot encoding (`handle_unknown='ignore'`) for categorical features. All transformers are fit on training folds only.

- **Model selection.** 5-fold cross-validation (shuffled, seed 42) on the training split only. The test set was used exactly once for final evaluation of all candidates.
- **Tuning.** `GridSearchCV` over Ridge α ∈ [10⁻3, 103] (13 log-spaced values), scored by negative CV RMSE.
- **Reproducibility.** Fixed seed 42 everywhere; recorded package versions (Appendix A); serialized final pipelines with `joblib` ; raw data unchanged.

4.2 Models and theory anchor

Ordinary Least Squares estimates coefficients by minimizing the sum of squared residuals:

$$\hat{\beta} = arg\ min_{\beta}\ \Sigma_i\ (y_i - \beta_0 - \beta^T x_i)^2 = (X^T X)^{-1} X^T y \quad (1)$$

Multiple Linear Regression is the scientific baseline and interpretability anchor of the study. Ridge adds an L2 penalty λΣβⱼZ to stabilize correlated coefficients; Lasso adds an L1 penalty λΣ|βⱼ| that can zero out coefficients; Elastic Net mixes both. Decision Trees capture nonlinearity and interactions but are high-variance; Random Forest reduces that variance by bagging; Gradient Boosting corrects errors sequentially. Predictive accuracy is reported as MAE and RMSE in original target units, plus the coefficient of determination:

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2 \quad (2)$$

4.3 Naive baselines

A mean-prediction baseline establishes whether any learning adds value. Test-set baseline RMSE is $87,619 (Ames), 1.145 in USD 100k units (California), and 12.95 in 10k-NTD/ping units (UCI, derived exactly from the reported Linear Regression RZ). Every fitted model beats its baseline by a wide margin on Ames and California; on UCI all models except the Decision Tree improve the baseline by roughly 37–56%.

5. Exploratory Data Analysis and Feature Engineering

EDA was performed on training data only; the test set was never used for feature-engineering decisions. Each dataset contributed distinct findings that materially influenced modeling.

5.1 Ames Housing

- **Right-skewed target.** Figure 1 shows the long right tail of sale prices. This justifies RMSE-sensitive evaluation and motivates the log-target extension discussed in Section 10.
- **Informative missingness.** Figure 2 shows that the most-missing features encode absence (no pool, no alley, no fence, no fireplace). Dropping these rows would discard ~90% of the data; they were encoded as explicit “None” categories instead.
- **Strong but correlated size/quality signals.** Figure 3 ranks the top numerical correlates of price — OverallQual (r = 0.79) and GrLivArea (r = 0.71) dominate, while garage size/count and basement/first-floor area are mutually redundant, motivating regularized models.

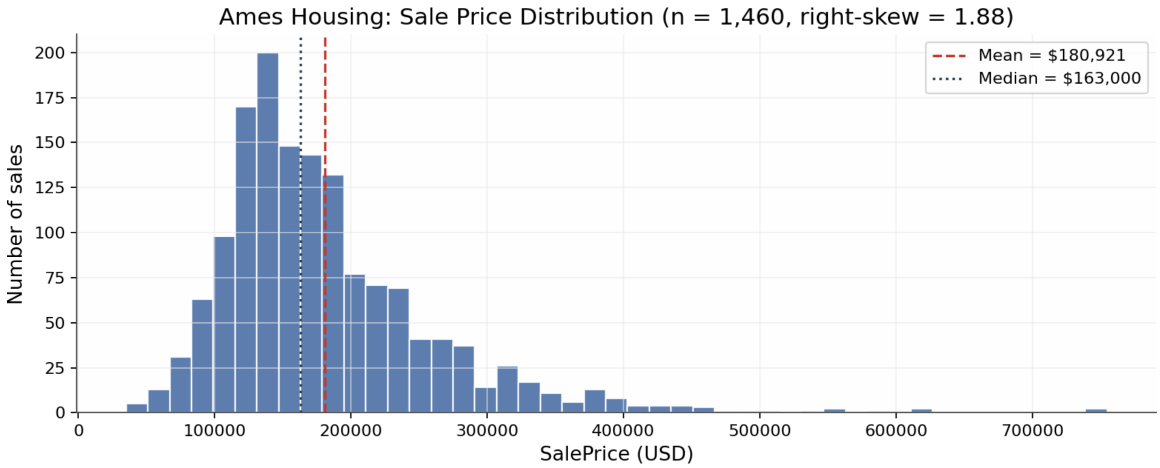

Figure 1. Ames Housing: sale-price distribution (n = 1,460). Right skew 1.88; mean exceeds median by ~$18k.

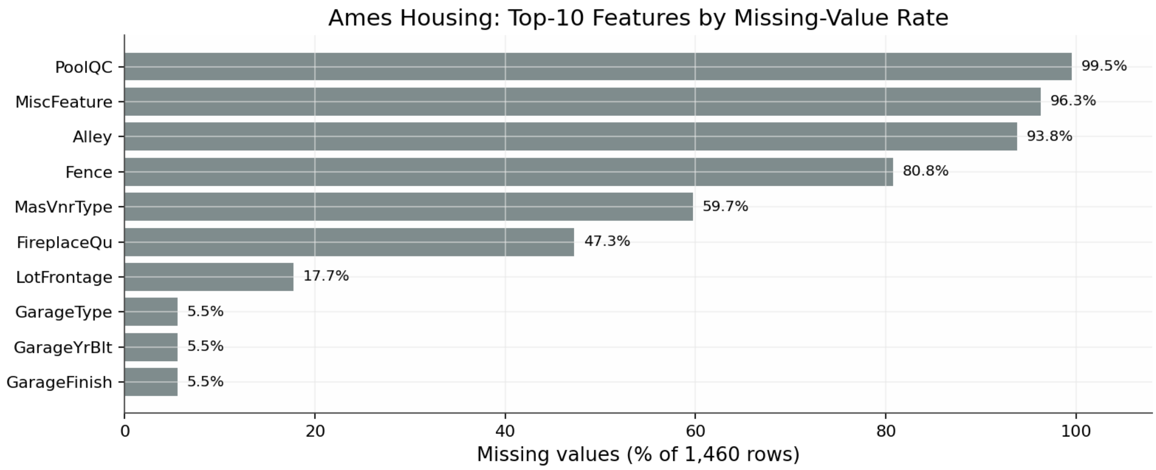

Figure 2. Ames Housing: top-10 features by missing-value rate. High rates reflect structural absence, not data corruption.

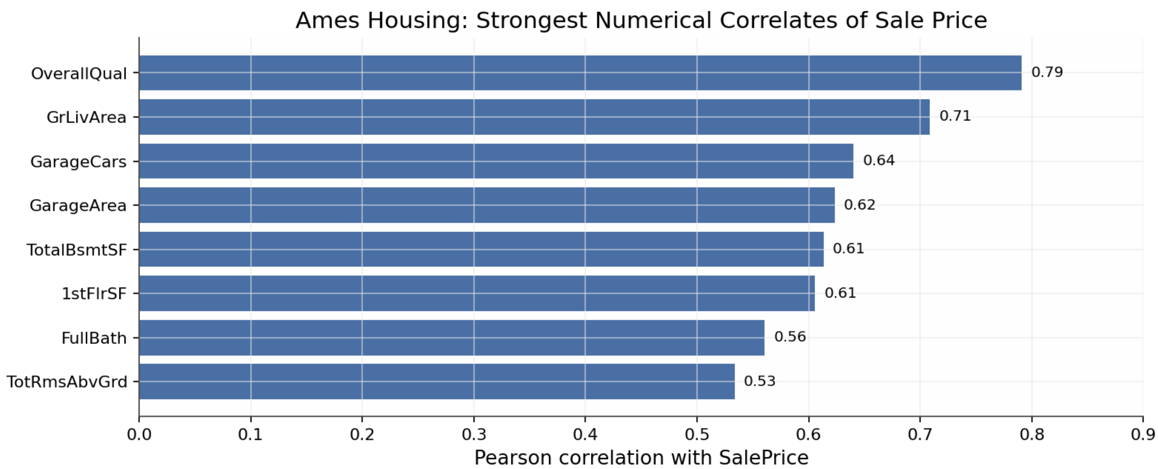

Figure 3. Ames Housing: strongest numerical correlates of SalePrice (Pearson r, full dataset).

5.2 California Housing

- **Right-censored target.** Figure 4 exposes a sharp spike at $500,000: 992 of 20,640 block groups (4.8%) sit at the cap, so the target is a *censored median*. RMSE at the top of the market is therefore understated, and no model can recover the true value of capped districts — a structural accuracy ceiling independent of model choice.
- **Income saturation.** Figure 5 (left) shows that value rises steeply with median income at low incomes and then flattens — a concave relationship with heteroscedastic spread that a single straight line must compromise on. This is the direct explanation for the linear family's RZ ceiling near 0.58 observed in Section 7.
- **Strong spatial structure.** Figure 5 (right) maps value by latitude/longitude: high values concentrate along the coast (Bay Area, Los Angeles/Orange County, San Diego) and fall sharply inland. Latitude and longitude interact — neither is informative alone — which is precisely the kind of pattern tree ensembles exploit and linear models cannot.

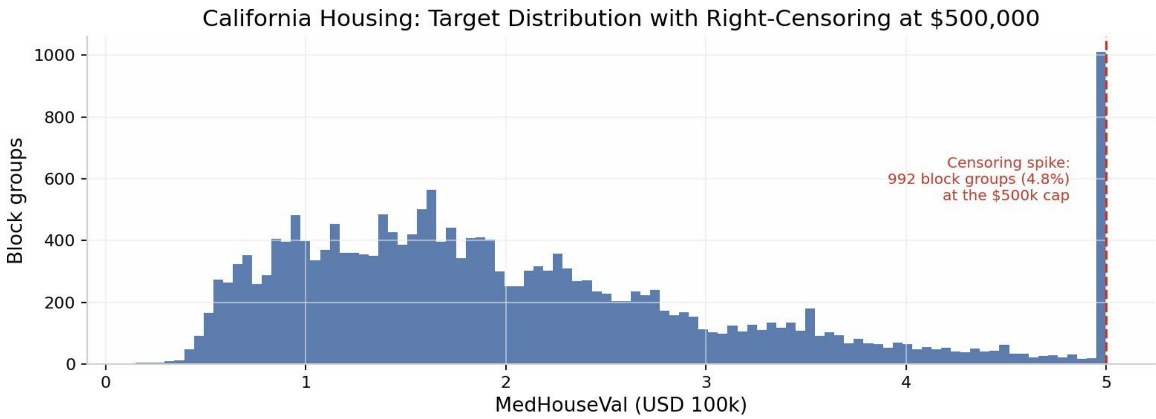

Figure 4. California Housing: target distribution. The spike at the right edge is right-censoring at the $500k cap (992 block groups, 4.8%).

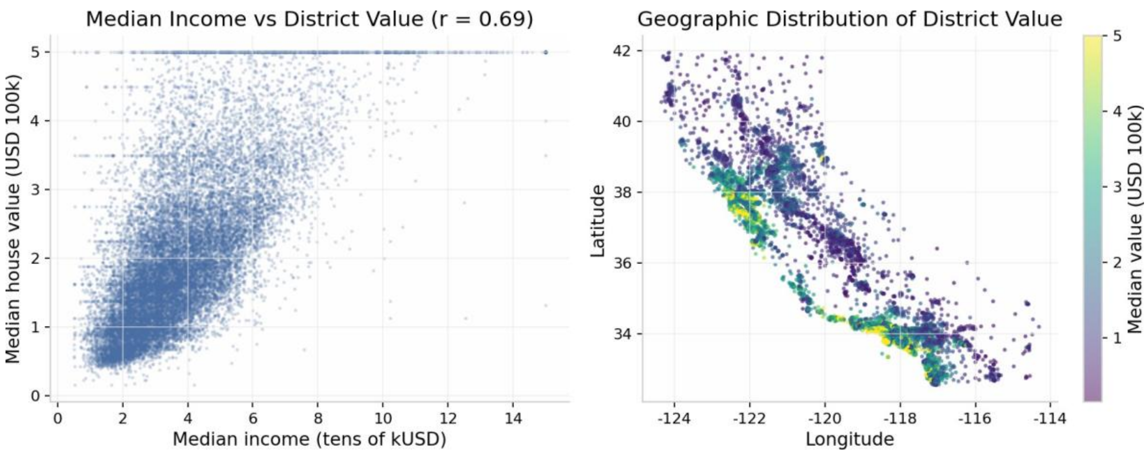

Figure 5. California Housing: median income vs district value showing saturation (left); geographic value surface showing coastal concentration (right).

5.3 UCI Real Estate Valuation

- **Log-linear transit premium.** Figure 6 (left) plots unit price against MRT-station distance on a log scale: price falls roughly linearly in log-distance (r = −0.67), the single strongest relationship in this dataset. A plain linear model on raw distance cannot represent this curvature; a log transform of distance is an obvious feature-engineering improvement.
- **Amenity and location effects.** Convenience-store count correlates positively with price (r = 0.57), with mean unit price rising from ~23 to ~43 (10k NTD/ping) as store count grows (Figure 6, right); latitude and longitude add further spatial signal (r ≈ 0.55 and 0.52).
- **Mild skew, no quality issues.** The target is only mildly right-skewed (skew 0.60; median 38.5, max 117.5), with zero missing values and no duplicates — consistent with this being the cleanest of the three datasets, and with the observation that all four linear-family models perform almost identically on it (Section 7).

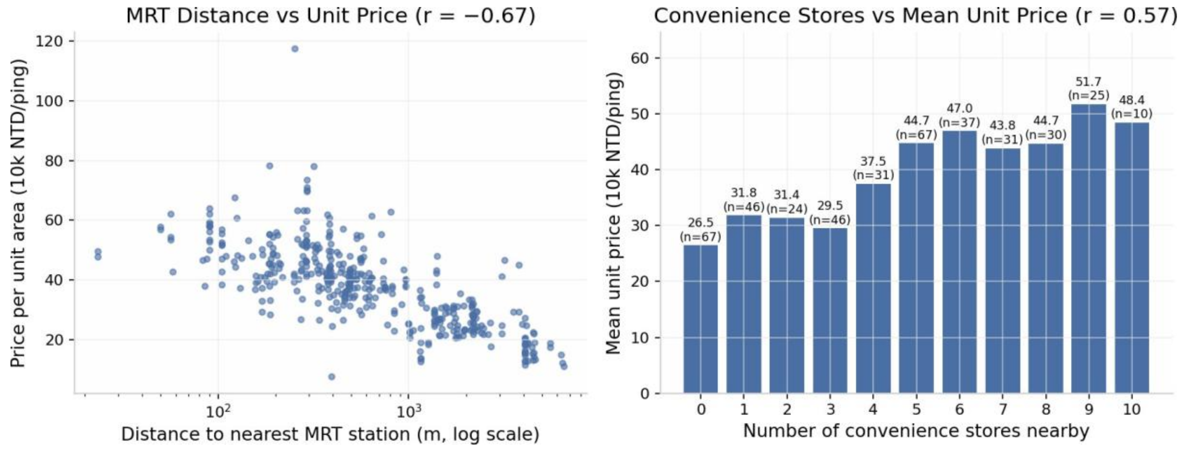

Figure 6. UCI Real Estate: unit price vs MRT-station distance on a log-distance scale (left); mean unit price by nearby convenience-store count (right).

6. Model Development

The experiment matrix followed the laboratory design: a dummy mean baseline (E0), a simple one-feature Linear Regression (E1: GrLivArea for Ames, MedInc for California, MRT distance for UCI), multiple Linear Regression through the full pipeline (E2), regularized variants (E3), a depth-limited Decision Tree and 300-tree Random Forest (E4), Gradient Boosting (E5), repetition on all three datasets (E6), and Ridge α tuning via grid search as the controlled tuning experiment (E7).

Key hyperparameters: Decision Tree `max_depth=6` ; Random Forest `n_estimators=300, min_samples_leaf=2` ; Gradient Boosting with scikit-learn defaults; Lasso/Elastic Net at α = 0.001 with `max_iter=20000` . All models shared the same preprocessing pipeline and split, so performance differences are attributable to the estimators rather than to data handling.

7. Evaluation Results

7.1 Ames Housing (target: SalePrice, USD)

Gradient Boosting was selected on cross-validated training performance (CV RMSE $27,709, best of seven) and confirmed on the held-out test set (Table 2, Figure 7). The linear family clusters tightly at RZ ≈ 0.88–0.90, while the single Decision Tree clearly underfits relative to ensembles.

Table 2. Ames Housing — test-set comparison (292 held-out sales) and 5-fold CV on training data

Model	Test MAE ($)	Test RMSE ($)	Test R^2	CV RMSE ($)	CV R^2
Gradient Boosting	**16,555.67**	**26,503.77**	**0.9084**	**27,708.84**	**0.8668**
Lasso	18,010.80	28,337.93	0.8953	36,042.14	0.7747
Random Forest	17,473.41	28,747.70	0.8923	30,637.61	0.8400
Linear Regression	18,287.70	29,473.87	0.8867	36,396.35	0.7700
Elastic Net	18,809.63	29,709.27	0.8849	33,588.56	0.8057
Ridge	19,004.41	29,841.82	0.8839	32,817.04	0.8147
Decision Tree	26,729.14	39,928.28	0.7922	44,010.85	0.6716

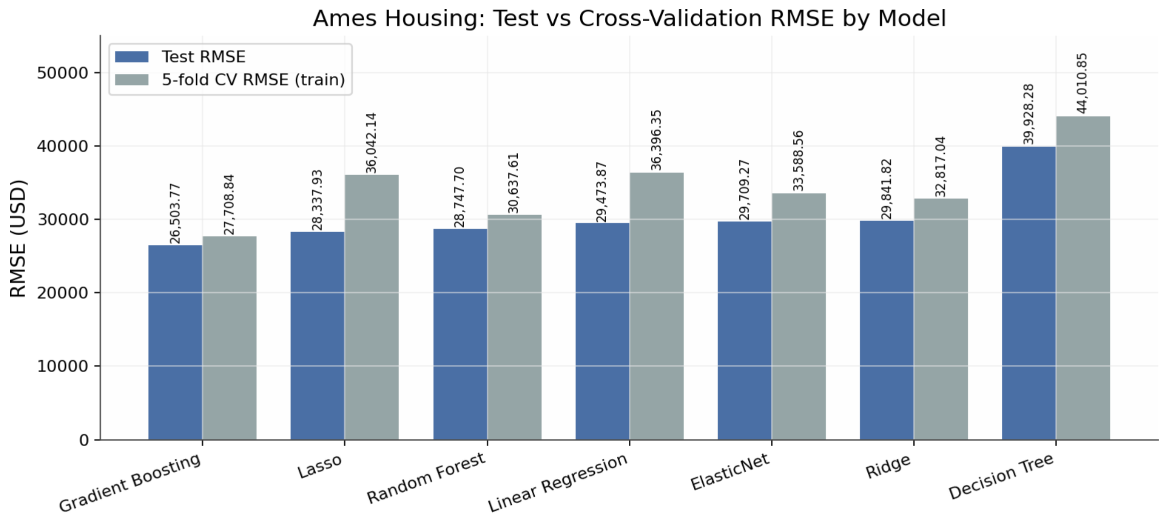

Figure 7. Ames Housing: test vs 5-fold CV RMSE. Gradient Boosting leads on both; the CV–test gap is small, indicating stable selection.

7.2 California Housing (target: median block-group value, USD 100k)

Random Forest was selected by CV and confirmed on test (Table 3, Figure 8). The striking finding is the linear family's collapse: Linear Regression and Ridge plateau at RZ ≈ 0.576, and Lasso at α = 0.001 shrinks the model to the mean (RZ ≈ 0.000), while ensembles exceed RZ 0.80 — direct evidence that the income–price and geography–price relationships are strongly nonlinear.

Table 3. California Housing — test-set comparison (4,128 held-out block groups) and 5-fold CV (units: USD 100,000)

Model	Test MAE	Test RMSE	Test R^2	CV RMSE	CV R^2
Random Forest	**0.3274**	**0.5051**	**0.8053**	**0.5109**	**0.8047**
Gradient Boosting	0.3717	0.5422	0.7756	0.5321	0.7881
Decision Tree	0.4539	0.7028	0.6230	0.7333	0.5973
Ridge	0.5332	0.7456	0.5758	0.7205	0.6115
Linear Regression	0.5332	0.7456	0.5758	0.7205	0.6115
Elastic Net	0.8060	1.0219	0.2031	1.0288	0.2080
Lasso	0.9061	1.1449	−0.0002	1.1562	−0.0002

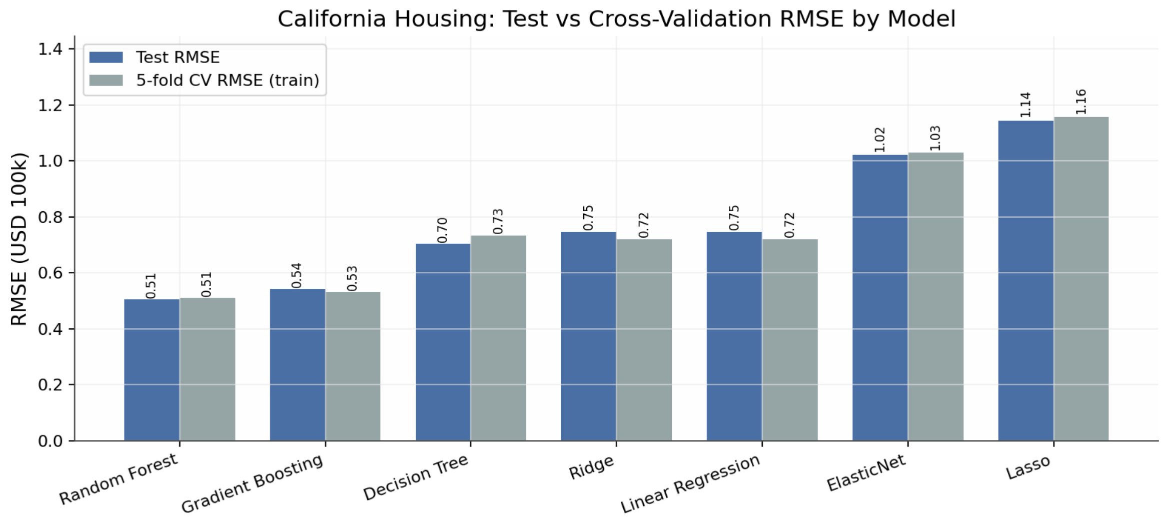

Figure 8. California Housing: test vs CV RMSE. Ensembles dominate; Lasso/Elastic Net collapse to the mean baseline at $\alpha = 0.001$.

7.3 UCI Real Estate Valuation (target: price per unit area, 10k NTD/ping)

Random Forest was again selected by CV (Table 4, Figure 9). Gradient Boosting achieved a marginally lower test MAE (3.904 vs. 3.913) but a higher test RMSE and clearly worse CV scores — under the stated selection rule (choose by cross-validated training performance, evaluate once on test), Random Forest remains the justified choice.

Table 4. UCI Real Estate — test-set comparison (83 held-out transactions) and 5-fold CV (units: 10,000 NTD per ping)

Model	Test MAE	Test RMSE	Test R²	CV RMSE	CV R²
Random Forest	**3.9131**	**5.6689**	**0.8084**	**7.8302**	**0.6643**
Gradient Boosting	3.9036	5.8443	0.7964	8.3502	0.6195
Ridge	5.3022	7.3108	0.6814	9.1083	0.5481
Elastic Net	5.3048	7.3139	0.6811	9.1105	0.5478
Lasso	5.3052	7.3144	0.6811	9.1108	0.5478
Linear Regression	5.3054	7.3148	0.6811	9.1109	0.5478
Decision Tree	5.9313	8.1530	0.6038	10.7388	0.3561

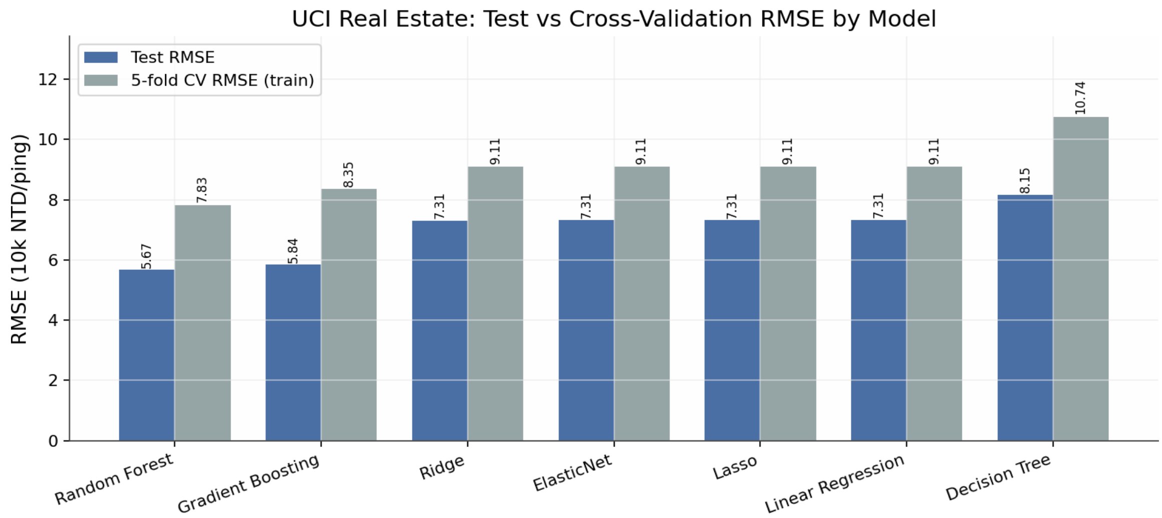

Figure 9. UCI Real Estate: test vs CV RMSE. The linear family is nearly identical across all four variants on this small dataset.

7.4 Cross-dataset synthesis

Figure 10 summarizes the selected models. Three consistent patterns emerge: (i) ensembles improve most where relationships are nonlinear and interactions matter (California: +0.23 RZ over linear; UCI: +0.13; Ames: +0.02); (ii) the CV–test gap is small on the two larger datasets, indicating stable selection; (iii) on the 414-row UCI dataset the test RZ (0.808) is notably higher than CV RZ (0.664), a small-sample artifact — the CV estimate is the more conservative and trustworthy expectation of future performance.

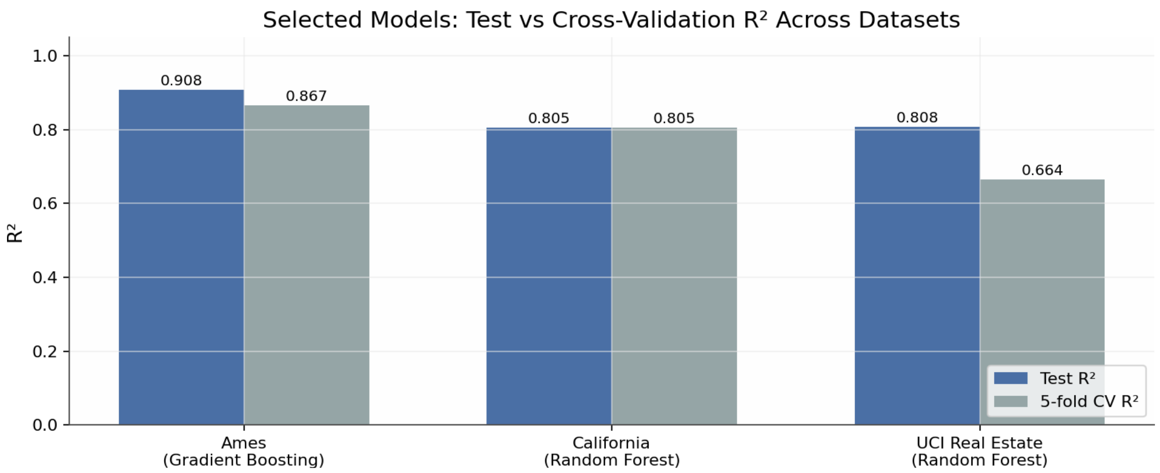

Figure 10. Selected models: test vs cross-validation RZ across the three datasets.

> **Adjusted R^2 caveat (Ames).** The reported adjusted RZ values for Ames are large negative numbers. This is an artifact, not a model failure: after one-hot encoding, the design matrix has more features (p) than test observations ($n = 292$), so the penalty term $(n-1)/(n-p-1)$ becomes negative. Adjusted RZ is only meaningful when $n \gg p$; cross-validation is the appropriate comparison tool in the high-dimensional regime.

8. Interpretation

8.1 What drives price

Findings are stated as conditional associations, not causal effects:

- **Ames.** Above-ground living area (GrLivArea), overall material/finish quality (OverallQual), neighborhood, and garage capacity dominate both linear coefficients and tree importances. The linear baseline's test RZ of 0.887 shows these attributes explain most of the price variance additively; Gradient Boosting's extra ~2 RZ points come from interactions (e.g., quality × area) and diminishing returns at the high end.
- **California.** Median income is by far the strongest signal, followed by location (latitude/longitude, i.e., coastal proximity), average rooms, and house age. The 23-point RZ gap between linear and ensemble models shows these effects are strongly nonlinear and interacting.
- **UCI.** Distance to the nearest MRT station, number of nearby convenience stores, geographic coordinates, and house age drive unit prices — a transit-accessibility premium consistent with urban economic intuition.

8.2 The linear baseline's role

Linear Regression never wins on accuracy, but it is far from useless. On Ames it is within ~$3,000 RMSE of the best model while remaining fully coefficient-interpretable; on UCI it matches the regularized variants almost exactly, confirming low multicollinearity there; and on California its failure is itself diagnostic, revealing how much structure is nonlinear. This is exactly the “interpretability anchor” role the study design intended.

8.3 Regularization behavior

Ridge never degraded materially below plain OLS and improved Ames CV RMSE by ~$3,600, showing mild variance reduction where one-hot categories create correlated columns. Lasso and Elastic Net behaved as expected on Ames and UCI but collapsed to the mean predictor on California at $\alpha = 0.001$ — a reminder that the useful regularization range is dataset-dependent and must be tuned by CV rather than assumed.

8.4 Generalization behavior

For both selected models the test metrics sit slightly better than or close to CV metrics (Ames: test RMSE $26,504 vs. CV $27,709; California: 0.505 vs. 0.511), so there is no evidence of overfitting at the selection stage. UCI shows the opposite direction (test better than CV), which is benign: it reflects fold-to-fold variance on 331 training rows rather than leakage, and argues for reporting CV uncertainty on small datasets.

9. Limitations and Risks

- **Historical drift.** All models are trained on past markets (Ames 2006–2010, California 1990, Taiwan 2012–2013). Inflation, interest-rate cycles, and market regime changes will degrade absolute price predictions; the models estimate relative structure better than current price levels.
- **Target semantics.** California predicts a *median block-group* value with censoring at $500,001, not an individual home price; UCI predicts price *per unit area*, not total price. Using either as an individual-property valuation would be a category error.
- **Random split realism.** A random holdout assumes deployment on similar properties from the same market and period. Production use would require temporal or spatial (grouped) splits to measure true forward-looking performance.
- **Small-sample uncertainty (UCI).** With 83 test rows, a handful of transactions moves RZ by several points; the CV estimate (0.664) should anchor expectations.
- **Fairness and governance.** Neighborhood and coordinate features can encode socioeconomic and historical inequalities. Use in credit, taxation, or access decisions requires governance review, segment-level error analysis, and human escalation rules for unusual properties.
- **Unmeasured drivers.** No dataset includes financing conditions, renovation history beyond year-remodeled, school quality, or micro-location attributes, capping attainable accuracy.
- **Limited tuning budget.** Only Ridge received a full grid search. The selected ensembles used fixed, sensible configurations; further tuning (learning rate, depth, subsampling) would likely close the remaining gap between CV leaders.

10. Future Improvements

- **Log-transformed target.** The Ames target skew (1.88) suggests `TransformedTargetRegressor` with log1p could reduce the influence of the high-end tail; final errors must be reported after inverse transformation.

- **Deeper tuning of ensembles.** Randomized/Bayesian search over Gradient Boosting and Random Forest hyperparameters, with a stated compute budget.
- **Spatial and temporal validation.** Neighborhood-grouped or year-based splits to measure realistic deployment performance; spatial features (distance-to-center) for California.
- **Uncertainty quantification.** Quantile regression or conformal prediction intervals so each estimate ships with an error range, as professional practice requires.
- **Segment-level error analysis.** Systematic evaluation by price band and neighborhood to detect where errors concentrate (e.g., very expensive or very old homes).
- **Deployment packaging.** Versioned artifacts (already saved via `joblib`), input-schema validation, drift monitoring, and periodic retraining.

11. Conclusion

The study met its stated success criteria on all three datasets. Every final model reduces RMSE by 56–70% relative to a naive mean baseline, generalizes stably (small CV–test gaps on the larger datasets), and was produced by a fully reproducible, leakage-safe pipeline with a fixed seed, recorded versions, and serialized artifacts. The central analytical question — how accurately, reliably, and transparently can property value be predicted — has a dataset-dependent answer: on the rich mixed-type Ames data, ensemble boosting reaches RZ = 0.908 with the linear baseline close behind and fully interpretable; on California and UCI, nonlinear structure is essential and tree ensembles are clearly superior. Linear Regression thus proves its dual value: a competitive, transparent baseline on additive problems, and a diagnostic instrument that exposes nonlinearity when it fails. Within their data coverage, the selected models are fit for decision support; beyond it — current markets, individual California homes, or high-stakes credit decisions — they require revalidation, uncertainty estimates, and human oversight.

Appendix A. Environment, Artifacts and Reproducibility

Table 5. Execution environment (from run_metadata.json; executed 2026-07-19)

Component	Version / setting	Component	Version / setting
Python	3.11.0	scikit-learn	1.8.0
numpy	1.26.4	matplotlib	3.10.8
pandas	2.3.3	seaborn	0.13.2
joblib	1.5.3	Random seed	42 (all splits, CV, stochastic models)

Table 6. Submitted artifacts per dataset

Dataset	Selected model	Model artifact (joblib)	Result CSVs
Ames Housing	Gradient Boosting	`ames_house_price_pipeline.joblib`	`ames_model_comparison.csv`, `ames_cross_validation.csv`
California Housing	Random Forest	`California_house_price_pipeline.joblib`	`California_model_comparison.csv`, `California_cross_validation_results.csv`
UCI Real Estate	Random Forest	`uci_real_estate_model.joblib`	`uci_model_comparison.csv`, `uci_cross_validation.csv`

Supporting files: three self-contained analysis notebooks (Ames, California, UCI), `23MID0043_Lab01_README.md` (execution order and environment), `execute_notebooks.py` (headless notebook re-execution harness), and `run_metadata.json` (seed, split sizes, selected models, headline metrics). Notebooks restart-and-run-all cleanly; raw data is never modified.

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